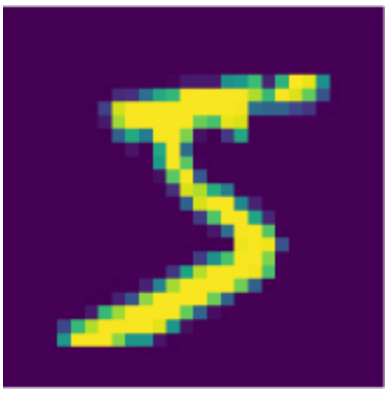
track gradients for, whereas `x` is an input, not a trainable parameter. We also got rid of the last activation function for reasons that will become clear. For this example, we will use the famous MNIST data set that contains images of handwritten digits from 0 to 9, such as the one in figure A.2.

Figure A.2. An example image from the MNIST dataset of hand-drawn digits.



We want to train our neural network to recognize these images and classify them as digits 0 through 9. PyTorch has a related library that lets us easily download this data set.

Listing A.4. Classifying MNIST using a neural network

```
mnist_data = TV.datasets.MNIST("MNIST", train=True, download=False)
1

lr = 0.001
epochs = 2000
batch_size = 100
lossfn = torch.nn.CrossEntropyLoss()
2
for i in range(epochs):
    rid =
np.random.randint(0,mnist_data.train_data.shape[0],size=batch_size)
3
    x = mnist_data.train_data[rid].float().flatten(start_dim=1)
4
    x /= x.max()
5
    pred = nn(x,w1,w2)
6
    target = mnist_data.train_labels[rid]
7
    loss = lossfn(pred,target)
```